1 Chain rule

Setup. Let $h : \mathbb{R}^n \to \mathbb{R}^m$ be one layer and $\ell : \mathbb{R}^m \to \mathbb{R}$ be everything after it (the rest of the network and the loss), so the loss is $L(x) = \ell\big(h(x)\big)$. The *local Jacobian* of the layer is the $m \times n$ matrix

$$J_h(x) = \frac{\partial h}{\partial x}, \qquad (J_h)_{ij} = \frac{\partial h_i}{\partial x_j}.$$

Chain rule.

$$\nabla_x L \;=\; J_h(x)^\top \, \nabla \ell\big(h(x)\big).$$

Here $\nabla \ell\big(h(x)\big)$ is the *downstream gradient* (what the rest of the network wants the layer's output to do), and $J_h(x)^\top$ converts it into a gradient on the layer's input x.

Intuition. The downstream gradient says how the loss would change if the layer's *output* moved. The local Jacobian says how the layer's output moves when its *input* x moves. Multiplying the two gives how the loss changes when x moves, which is what gradient descent needs to update x. Backpropagation repeats this layer by layer, so every layer's local Jacobian is a factor in the gradient that reaches earlier layers.

Two cases relevant to this paper.

1. **Local Jacobian is zero**, $J_h(x) = 0$. Then

   $$\nabla_x L = 0 \quad \text{for every downstream gradient.}$$

   Meaning: small changes to x do not change the layer's output, so the loss gives x no direction to move. Gradient descent never updates x, and nothing before this layer learns either. This is what rounding (quantization) does.

2. **Local Jacobian is the identity**, $J_h(x) = I$. Then

   $$\nabla_x L = \nabla \ell\big(h(x)\big).$$

   Meaning: the downstream gradient is passed back unchanged, as if the layer did nothing. Whatever the layer actually did to x (for example, adding a rounding error) has no effect on the gradient, so earlier layers get no signal about it. This is what the Straight-Through Estimator (STE) assumes.

2 Ridge regression

Why needed. The paper's dequantizer g is a ridge-regression fit of the original x from the rounded q (with $u = q$, $v = x$ below).

Notation. For $u, v \in \mathbb{R}^N$: $\bar{u} = \frac{1}{N}\sum_i u_i$, $\overline{uv} = \frac{1}{N}\sum_i u_i v_i$, $\mathrm{Var}_u = \overline{u^2} - \bar{u}^2$, $\mathrm{Cov}_{uv} = \overline{uv} - \bar{u}\,\bar{v}$, and $\mathbf{1}$ is the all-ones vector.

Problem. Fit $v \approx a\,u + c\mathbf{1}$, penalizing only the slope:

$$\min_{a,c} \frac{1}{2N}\|a\,u + c\mathbf{1} - v\|^2 + \frac{\lambda}{2}a^2, \qquad \lambda > 0.$$

Setting derivatives to zero:

$$c^* = \bar{v} - a^*\bar{u}, \qquad a^* = \frac{\mathrm{Cov}_{uv}}{\mathrm{Var}_u + \lambda}.$$

Without intercept ($c = 0$): $a^* = \dfrac{\overline{uv}}{\overline{u^2} + \lambda}$.

Fit. The vector of the line's predictions at the data points:

$$\text{fit} = a^*u + c^*\mathbf{1} = a^*(u - \bar{u}\mathbf{1}) + \bar{v}\mathbf{1}.$$

Role of λ. (i) *Stability:* the denominator is at least $\lambda > 0$, so no division by zero even if all u_i are equal. (ii) *Shrinkage:* as $\lambda \to \infty$, $a^* \to 0$ and the fit becomes the mean $\bar{v}\mathbf{1}$.